

ELEC 1100 Laboratory 3: Summary Sheet

Name: _____

Student ID:

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Notes:

- (1) Show demo to your TA for an attendance record; **bonus mark** will be given to **groups** who can complete all TA Checks in the first 2 hours of your enrolled lab session.

LA2 (Fri, before 15:30), LA3 (Mon, before 12:30), LA1 (Tue, before 15:30)

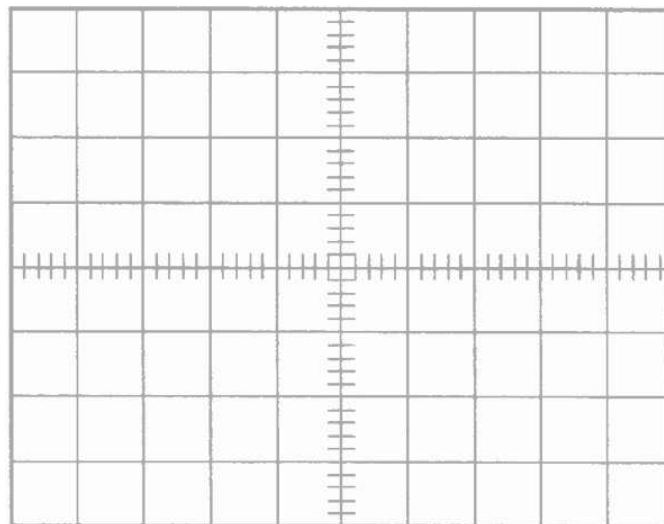
- (2) Individually submit your completed lab summary sheet to Canvas (LA1/LA2/LA3) "Assignments" before the deadline.

LA2 (Fri, deadline 16:20), LA3 (Mon, deadline 13:20), LA1 (Tue, deadline 16:20)

Experimental Results and Analysis

Q1: Sketch what you see on the DSO (both CH1 & CH2 waves, including the voltage & time scales and zero references displayed on the screen).

Notice: Do the sketching on the grid below by your hand, photos of DSO screen will NOT be accepted for grading.



Q2: What is the difference between the two waveforms? Briefly explain your answer (Explain the feature of a diode).

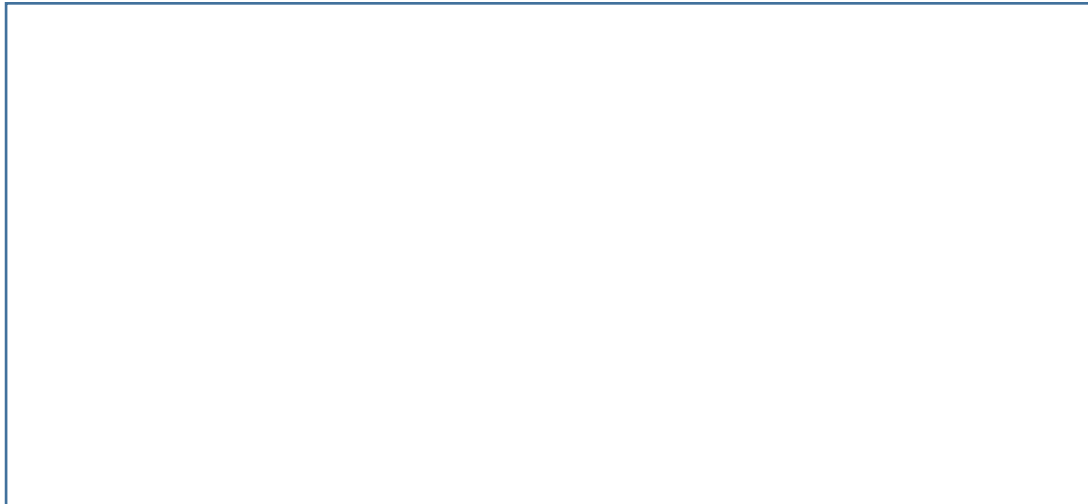
Q3: Complete the table of experiment results (2-resistor circuit).

V_{in} (V)	1	2	3	4	5	6	7	8
V_Z (V)								
V_{in} (V)	9	10	11	12	13	14	15	16
V_Z (V)								

Q4: Complete the table of experiment results (Zener diode circuit).

V_{in} (V)	1	2	3	4	5	6	7	8
V_Z (V)								
V_{in} (V)	9	10	11	12	13	14	15	16
V_Z (V)								

Q5: Use the data you record in Q3 and Q4 to make a **line chart** of V_Z against V_{in} and fill in the space provided below.



Q6: From your record in Q4, determine the breakdown voltage (give a rough number) of the Zener diode.

Breakdown voltage = _____

Q7: What is the multimeter reading (V_{out}) displayed when $V_{in} = 8V$? _____

Q8: What is the multimeter reading (V_{out}) displayed when $V_{in} = 12V$? _____

Q9: Use the results in Q7 and Q8, determine the line regulation of your LM7805 voltage regulator (Show your calculation).

$$\text{Line regulation} = \frac{\Delta V_O}{\Delta V_I}$$